

THE UNIVERSITY OF LAHORE

Department of Computer Science & IT

Faculty of Information Technology

Final Year Project Proposal

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PRODUCT TITLE:	AI-Based Real-Time Weapon Detection System Using Thermal and RGB Image Fusion
KEY WORDS:	Artificial Intelligence, Computer Vision, YOLOv8, Thermal Imaging, RGB Fusion, Weapon Detection, Object Detection, Surveillance System
DOMAIN OF THE PRODUCT:	Artificial Intelligence, Computer Vision, Embedded Systems, Security & Surveillance
SUPERVISOR'S NAME:	Temur ul Hassan

STUDENT INFORMATION <i>Write in BLOCK LETTERS ONLY.</i>				
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Please expand space according to answer

1- Problem Identification:

Identify a real-world problem or a market gap that can be addressed through your project. This is the foundation of any entrepreneurial venture.

In recent years, security threats involving weapons such as guns and knives have increased, creating a need for intelligent surveillance systems. Existing weapon detection systems mainly rely on RGB cameras, which perform poorly in low-light, foggy, or night-time environments. Additionally, many systems depend on expensive infrastructure and cloud-based processing, making them inaccessible for small-scale use.

There is a clear need for a **cost-effective, real-time weapon detection system** that can operate reliably in all environmental conditions. This project addresses this gap by integrating **thermal imaging with RGB cameras**, enabling detection even in darkness or visually challenging scenarios.

1- Product Market Analysis:

Conduct thorough market research to understand the target audience, competitors, and potential customers for your proposed solution.

The market for AI-based surveillance systems is rapidly growing, especially in areas such as:

- Smart cities
- Security agencies
- Airports and public places
- Educational institutions

Existing solutions like enterprise surveillance systems are expensive and require advanced infrastructure. Research-based solutions using YOLO models exist but typically rely only on RGB input and lack robustness in low-light conditions.

The target users for this system include:

- Security organizations

- Government agencies
- Small institutions (schools, offices)
- Private security firms

This project fills the gap by providing a **low-cost, intelligent, and adaptive detection system**.

2- Unique Value Proposition:

Define the unique features or advantages of your project that differentiate it from existing solutions in the market. This will help you to understand your competitive advantage

The proposed system offers several unique advantages:

- Combines **thermal + RGB image fusion**
- Works effectively in **low light, night, and harsh environments**
- Provides **higher detection accuracy** compared to RGB-only systems
- Does not rely heavily on cloud → **better privacy and faster response**
- Can be implemented on **low-cost hardware (Jetson Nano / Raspberry Pi)**

This makes the system more reliable, practical, and suitable for real-world deployment.

3- Features:

Write the overall features to solve problem.

- Real-time weapon detection (guns, knives)
- Dual camera input (RGB + Thermal sensor)
- Image fusion module for enhanced detection
- YOLOv8-based object detection model
- Bounding box detection with labels
- Alert system (visual + alarm notification)
- Works in low-light and night conditions
- Edge-based processing (low latency)
- Scalable for integration with CCTV systems

4- Revenue Generation:

Explore different revenue streams for your project, such as subscriptions, advertising, or partnerships. This will help you to understand the financial viability of your entrepreneurial venture.

The project has strong potential for commercialization through:

- Selling the system as a **security product**
- Subscription-based monitoring services
- Partnerships with security companies
- Government and institutional deployment contracts
- Integration with existing CCTV systems as a paid upgrade

5- Marketing and Sales Strategy:

Develop a plan for marketing and selling your product or service. This includes identifying target marketing channels, customer acquisition strategies, and pricing models.

- Target security companies and institutions
- Demonstrate working prototype to stakeholders
- Use digital marketing (websites, LinkedIn, tech platforms)
- Collaborate with hardware vendors
- Offer pilot deployments for small organizations
- Pricing strategy: affordable compared to enterprise systems

6- Risk Assessment:

Identify potential risks and challenges associated with your project. This can include technical, market, or financial risks to develop mitigation strategies.

Technical Risks:

- Difficulty in aligning RGB and thermal images
- Hardware integration challenges

Mitigation: Use pre-aligned datasets and tested sensors

Financial Risks:

- Thermal cameras can be expensive

Mitigation: Use low-cost sensors or simulated datasets

Performance Risks:

- Model accuracy issues in real environments

Mitigation: Train on diverse datasets and test extensively

7- Scalability and Growth Potential:

Analyse the scalability and growth potential of your project. This involves considering factors such as market demand, scalability of technology, and expansion opportunities.

The system has strong scalability potential:

- Can be integrated into **smart city infrastructure**
- Expandable to detect more objects (suspicious behavior, faces)
- Can support **multi-camera systems**
- Deployable in airports, malls, and defense sectors
- Future integration with IoT and cloud analytics

8- Implementation and Execution Plan:

Develop a detailed plan for implementing and executing your project. This can include milestones, timelines, resource allocation, and team roles. Remember, the ultimate goal is to understand how your FYP can be transformed into a viable and sustainable entrepreneurial venture.

The project will be completed in the following phases:

- 1) **Requirement Analysis & Design:** Define system architecture, select hardware (RGB + thermal camera), and choose software tools.
- 2) **Data Collection & Preparation:** Gather RGB and thermal weapon datasets, perform preprocessing and augmentation.
- 3) **Image Fusion:** Align and combine RGB and thermal images using suitable fusion techniques.
- 4) **Model Training:** Train a YOLOv8 model on the fused dataset and evaluate using accuracy metrics.
- 5) **System Integration:** Integrate cameras, fusion module, and detection model for real-time operation.
- 6) **Alert & Interface Development:** Develop a display system with detection output and alert mechanism.
- 7) **Testing & Deployment:** Test under different conditions and finalize the working prototype.

Supervisor's Signature:

Suggested Improvements regarding FYP (*if any*):
